HERIOT WATT UNIVERSITY

School of Mathematical and Computer Sciences

Actuarial Mathematics and Statistics

F70LB

Life Insurance Mathematics B

Excel Test – SAMPLE PAPER

April Examination and Assessment Diet 2024-2025

Duration: 2 Hours 30 minutes

Please answer all questions in Part A and Part B of this paper.

Total 90 marks

Important Instructions – PLEASE READ

You have been supplied with two answer booklets in which to complete your work. One is an Excel spreadsheet and the other is a Word document. Note that Part B is a written exercise which depends on Part A, so you will need to attempt Part A before proceeding to Part B.

Submit your answers to Canvas as two files:

- *Part A: Completed "Part A Excel Answer Booklet.xlsx"*
- *Part B: Completed "Part B Word Answer Booklet.docx"*

Mark scheme:

- *Part A (two questions **65 marks**): Marks are awarded for correct answers and working. There are no marks directly awarded for how you set out your spreadsheets. However, it is more likely to be possible to award marks for working if your spreadsheet is clear and easy to follow.*
- *Part B (one question **25 marks**): In part B, you will be asked to write about your answers from Part A.*

Ethical Commitment: By submitting answers to this paper, I declare that these answers are entirely my own work and have not been shared, in part or in whole or in any form, with any other student, or disseminated in any way.

I also confirm that I have reviewed and understand all University regulations, policies and procedures on assessment misconduct, including but not limited to accessing unauthorised materials during this examination or the use of unauthorised electronic devices.

I understand that infringing this statement would represent an academic misconduct offence subject to disciplinary action according to the University Regulations and Policy regarding Academic Misconduct, with possible significant consequences for degree progression and final degree outcome.

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Part A

Question 1

A life insurance company issues a 25-year term assurance policy to a life aged 32 exact. The worksheet 'Q1 Base' gives details of the premiums charged and benefits payable, as well as the reserving basis used.

Calculate the gross premium policy value (reserve) at the end of each policy year in two ways:

(i) By direct calculation of the prospective policy value without use of a recursive relationship between policy values

(ii) By using a recursive relationship starting from $V(0) = 0$ and working forwards in time.

The calculation in part (ii) is a calculation of the *retrospective* policy value (reserve). In this question, you should not expect the prospective and retrospective reserves to be equal.

HINT: In part (i), you can use the following formula where S is the sum assured, E is the annual initial expense, α is the percentage of each premium paid as an expense, and P is the annual premium. You will need to start by calculating the assurance and annuity factors.

$$V(t) = SA^{1}_{x+t:\overline{n-t}|} + (E + \alpha P)\ddot{a}_{x+t:\overline{n-t}|} - P\ddot{a}_{x+t:\overline{n-t}|}$$

[Total 23 marks]

PLEASE TURN OVER

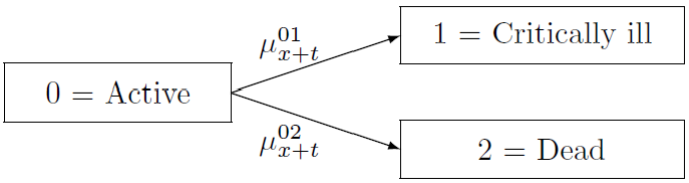
Question 2

A life insurance company offers 20-year unit-linked endowment assurances to lives aged 30 exact. Premiums of £2,000 a year are payable annually in advance for the term of the policy or until payment of any benefit if earlier.

The death benefit, payable at the end of the year of death, is the greater of £15,000 and the value of the units. On maturity, the value of the units is payable.
The assurance includes an additional critical illness benefit. On diagnosis of a critical illness, the value of the units is payable immediately and all cover ceases.

The level allocation rate is 95%.

The company uses the following model:



The remaining policy terms and profit test assumptions, including transition intensities for the model are available on the 'Q2 Base' worksheet.

(i) Calculate the unit fund value at the end of each policy year.

(7 marks)

(ii) Calculate using an Euler scheme:

a. The conditional probabilities ${}_1p_{x+(t-1)}^{01}$ and ${}_1p_{x+(t-1)}^{02}$ of critical illness and death in the next year given that life is active at time $t-1$ for each $t = 1,2,\ldots,20$.

(9 marks)

b. The probability that the maturity benefit is paid.

(2 marks)

The company decides to set a new level allocation rate such that the present value of profits is equal to 10% of the annual premium. All other policy terms and assumptions remain unchanged.

(iii) Project the unit and non-unit funds and calculate the present value of profits and premiums. Hence determine the new allocation rate.

(24 marks)

[Total 42 marks]

PLEASE TURN OVER

Part B

Question 1

You are in an actuarial graduate role and you have have spent your afternoon working on the calculations in Part A. You showed the calculations to your junior colleague, Xan, who has just joined the company. Xan has a Maths degree but is new to actuarial work. You are not going to be at work tomorrow, so before you go home today you have decided to send an email to Xan to follow up on some points that came up when you discussed.

You are comfortable that Xan understood your explanation of the spreadsheet when you discussed earlier, so you do not need to explain in general what the calculations are for or how they work.

In your email, you should include:

- A brief explanation of why the prospective and retrospective reserves are not equal for the policy in Part A Question 1.

  (4 marks)

- Whether the company should hold non-unit reserves for the contract in Part A Question 2. You should not do any further calculations to help you with this.

  (3 marks)

- Briefly, some of the most important things that should be done to make your spreadsheets comply with best Excel practice and why they matter. You can write about what you have actually done, what more should be done, or both.

  (8 marks)

A further 10 marks are available for writing and structure of your email.

You should write no more than 500 words.

[Total 25 marks]

END OF PAPER